

CONTROLLING 3-PHASE INDUCTION MOTORS Using VFD and PLC



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Various automation processes in the industry need control of AC induction motors using AC drives. Presented here is a robust system for switching on/off, varying the speed and direction of rotation of an industrial 3-phase induction motor. We use here Delta AC motor drive for its operation.

A simple control panel is wired using an Allen Bradley PLC for demonstration. An extended Intouch wonderware SCADA can be developed similar to our earlier article published in May 2015 issue for a PC based virtual control panel.

An electrical motor is an electromechanical device that converts electrical energy into mechanical energy. In case of 3-phase AC operation, the most-widely-used motor is the 3-phase induction motor as this type of motor does not require any starting device, being a self-starting motor.

What drives are

Often in the industry, need arises for controlling the speed of a 3-phase induction motor. Delta's AC motor drives are able to efficiently control motor speed, improve machine automation and save energy. Each drive in its variable frequency drive (VFD) series is designed to meet specific application needs.

AC drives accurately control torque, smoothly handle increased load and provide numerous custom control and configuration operating modes. A VFD can be used to vary speed, direction and other parameters of a 3-phase motor. We use the 2-wire method for controlling the speed and direction of the motor.

Working of a VFD

The first stage of a VFD is the converter, which comprises six diodes, which are

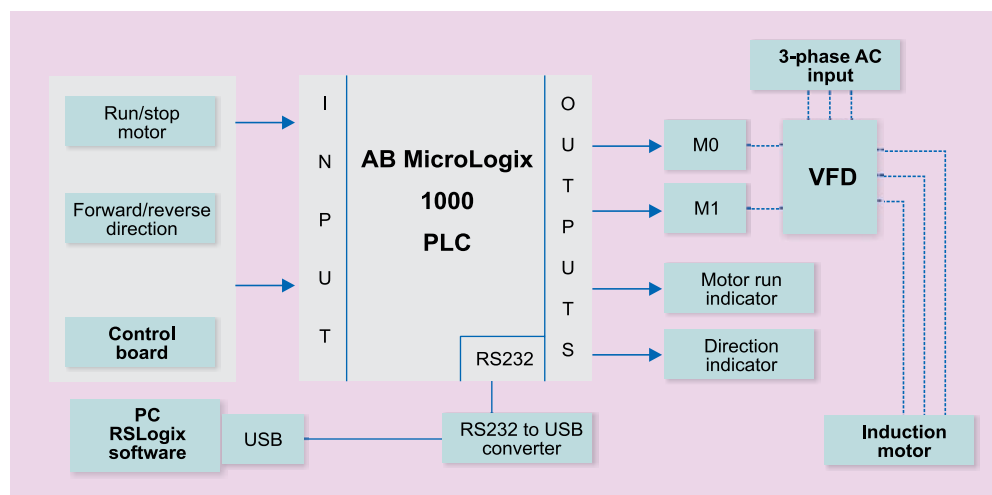


Fig. 1: Entire system layout

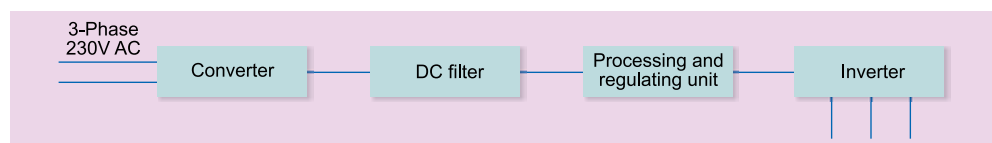


Fig. 2: Block diagram of a drive